1. Which idealization assumes that the dimensions of a body are negligible compared to other dimensions?

(A) Particle
(B) Rigid body
(C) Deformable body
(D) Continuum

2. A body is defined as rigid in mechanics if:

(A) Deformations are completely neglected under loads
(B) It deforms plastically under all loads
(C) It has zero mass and zero volume
(D) It fractures instantly under tension

3. According to Newton's First Law of Motion, a body remains at rest or moves with uniform velocity unless acted upon by:

(A) An unbalanced external resultant force
(B) An internal balanced stress system
(C) A gravitational couple only
(D) A frictional resistance

4. The principle of transmissibility states that a force acting on a rigid body can be transmitted to any point along its:

(A) Line of action without changing external effect
(B) Line of action without changing internal stress
(C) Perpendicular bisector
(D) Direction of motion only

5. The principle of transmissibility cannot be used to determine:

(A) Internal stress and deformation distributions
(B) External support reactions
(C) Overall static equilibrium
(D) Resultant of concurrent forces

6. Which of the following is a fundamental scalar quantity in classical engineering mechanics?

(A) Mass
(B) Force vector
(C) Linear momentum
(D) Torque

7. If two concurrent forces P and Q act at an angle theta, the magnitude of their resultant R is given by:

(A) sqrt(P^2 + Q^2 + 2*P*Q*cos(theta))
(B) sqrt(P^2 + Q^2 - 2*P*Q*cos(theta))
(C) P + Q + 2*P*Q*cos(theta)
(D) sqrt(P^2 + Q^2)

8. If two equal forces P act at an angle of 90 degrees to each other, their resultant magnitude is:

(A) P * sqrt(2)
(B) 2 * P
(C) P / sqrt(2)
(D) P / 2

9. If two forces of magnitude P act at an angle of 120 degrees, their resultant magnitude equals:

(A) P
(B) 2 * P
(C) P * sqrt(3)
(D) P / 2

10. If two forces P and Q act in opposite directions along the same line of action (theta = 180 deg), the resultant is:

(A) |P - Q|
(B) P + Q
(C) sqrt(P^2 + Q^2)
(D) 0 always

11. The angle alpha that resultant R makes with force P when P and Q act at angle theta is:

(A) atan((Q*sin(theta)) / (P + Q*cos(theta)))
(B) atan((P*sin(theta)) / (Q + P*cos(theta)))
(C) atan((Q*cos(theta)) / (P + Q*sin(theta)))
(D) atan(P / Q)

12. The Triangle Law of Forces states that if two forces are represented by two adjacent sides of a triangle in order, their resultant is:

(A) The third side taken in reverse order
(B) The third side taken in the same order
(C) The altitude of the triangle
(D) The centroid of the triangle

13. The Polygon Law of Forces is a graphical representation for determining the resultant of:

(A) Any coplanar concurrent force system
(B) Only three parallel forces
(C) Only non-coplanar couples
(D) Distributed loads exclusively

14. For a system of coplanar concurrent forces to be in static equilibrium, the necessary conditions are:

(A) Sigma Fx = 0 and Sigma Fy = 0
(B) Sigma Fx = 0 and Sigma M = 0 only
(C) Sigma Fy = 0 and Sigma M = 0 only
(D) Sigma Fx + Sigma Fy = 1

15. Lami's Theorem is applicable strictly to a system of:

(A) Three coplanar concurrent forces in equilibrium
(B) Any number of concurrent forces
(C) Three parallel forces in equilibrium
(D) Non-coplanar couples

16. Mathematically, Lami's Theorem is expressed as:

(A) P/sin(alpha) = Q/sin(beta) = R/sin(gamma)
(B) P/cos(alpha) = Q/cos(beta) = R/cos(gamma)
(C) P*sin(alpha) = Q*sin(beta) = R*sin(gamma)
(D) P + Q + R = 0

17. In Lami's theorem, the angle alpha in the denominator of P is the angle between:

(A) Forces Q and R
(B) Forces P and Q
(C) Forces P and R
(D) The resultant and force P

18. A Free Body Diagram (FBD) is defined as a diagram showing:

(A) A body isolated with all active and reactive forces
(B) Only internal stress resultants
(C) The deformed geometry of the structure
(D) The structural material microstructure

19. The moment of a force about an axis or point represents its tendency to produce:

(A) Rotation about that point or axis
(B) Pure linear translation along the line of action
(C) Volume change of the rigid body
(D) Plastic shear slip

20. In SI units, the moment of a force is expressed in:

(A) Newton-metre (N-m)
(B) Newton per metre (N/m)
(C) Joule per second (W)
(D) Pascal (Pa)

21. In vector mechanics, the moment of a force F about origin O with position vector r is given by:

(A) M_O = r x F
(B) M_O = F x r
(C) M_O = r . F
(D) M_O = (r . F) / |r|

22. Varignon's Theorem (Principle of Moments) states that:

(A) Moment of resultant equals sum of moments of components
(B) Moment of resultant is always zero
(C) Sum of forces equals sum of moments
(D) Couples cannot be replaced by single forces

23. A couple consists of two forces that are:

(A) Equal in magnitude, opposite in direction, non-collinear
(B) Equal in magnitude, same direction, collinear
(C) Unequal in magnitude, opposite in direction
(D) Perpendicular to each other at a common point

24. The resultant linear force of a couple is always:

(A) Zero
(B) Twice the magnitude of either force
(C) Equal to the moment arm distance
(D) Dependent on the reference origin

25. The moment of a couple is classified as a:

(A) Free vector
(B) Sliding vector
(C) Fixed bound vector
(D) Null vector

26. A frictionless roller support on a horizontal plane provides:

(A) One vertical reaction force
(B) One horizontal reaction force
(C) Two orthogonal reaction forces
(D) One vertical force and one reactive moment

27. A frictionless pin or hinged support in a 2D plane provides:

(A) Two independent orthogonal reaction forces
(B) One single reaction force normal to the pin
(C) Three independent reaction components
(D) A reactive couple only

28. A fixed (built-in or encastre) support in a 2D plane provides:

(A) Two reaction forces and one restraint moment (3 unknowns)
(B) Two reaction forces only
(C) One normal reaction force only
(D) One restraint couple only

29. A beam supported by a pin at one end and a roller at the other end is termed a:

(A) Simply supported beam
(B) Cantilever beam
(C) Continuous beam
(D) Propped cantilever beam

30. A beam supported fixed at one end and completely unsupported (free) at the other is a:

(A) Cantilever beam
(B) Overhanging beam
(C) Fixed-ended beam
(D) Simply supported beam

31. For static determinacy of a general coplanar structure using 3 equations, the number of unknowns must be:

(A) Equal to 3
(B) Greater than 3
(C) Equal to 6
(D) Equal to 1

32. The total equivalent concentrated point load for a uniformly distributed load (UDL) of w over length L is:

(A) w * L
(B) w * L / 2
(C) w * L^2 / 2
(D) 2 * w * L

33. The line of action of the equivalent concentrated load of a uniform rectangular UDL acts at:

(A) L / 2 from either end
(B) L / 3 from the left end
(C) 2*L / 3 from the left end
(D) L / 4 from the end

34. The equivalent concentrated point load for a triangular distributed load (UVL) of peak intensity w over length L is:

(A) (1/2) * w * L
(B) w * L
(C) (1/3) * w * L
(D) (2/3) * w * L

35. The line of action of the equivalent point load for a triangular UVL with apex at x=0 and base at x=L acts at:

(A) 2*L / 3 from the apex (1/3 from the base)
(B) L / 2 from the apex
(C) L / 3 from the apex
(D) 3*L / 4 from the apex

36. A force system where lines of action of all forces lie in the same plane is termed:

(A) Coplanar force system
(B) Collinear force system
(C) Concurrent 3D force system
(D) Spatial non-coplanar system

37. A force system where lines of action of all forces pass through a single common point is termed:

(A) Concurrent force system
(B) Parallel force system
(C) Coplanar non-concurrent system
(D) Couple system

38. If the algebraic sum of resolved forces along two perpendicular axes is zero (Sigma Fx=0, Sigma Fy=0), the system:

(A) Has zero resultant force
(B) Has zero resultant moment always
(C) Must be in dynamic acceleration
(D) Cannot produce a couple

39. Two equal and opposite parallel forces F separated by perpendicular distance d produce a moment equal to:

(A) F * d
(B) 2 * F * d
(C) F * d / 2
(D) F^2 * d

40. The scalar product (dot product) of two perpendicular force vectors is:

(A) 0
(B) 1
(C) Product of their magnitudes
(D) -1

41. Two forces of 30 N and 40 N act at right angles (theta = 90 deg). What is their resultant magnitude?

(A) 50 N
(B) 70 N
(C) 10 N
(D) 35 N

42. Two forces of 100 N and 100 N act at an angle of 60 degrees. What is their resultant magnitude?

(A) 173.2 N
(B) 141.4 N
(C) 100.0 N
(D) 200.0 N

43. Two forces of 60 N and 80 N act at an angle of 60 degrees. What is their resultant magnitude?

(A) 121.7 N
(B) 140.0 N
(C) 100.0 N
(D) 110.5 N

44. The resultant of two forces 5 N and 12 N acting at 90 degrees makes what angle with the 12 N force?

(A) 22.62 deg
(B) 67.38 deg
(C) 45.00 deg
(D) 30.00 deg

45. A force F = 200 N is directed at an angle of 30 degrees above the positive x-axis. Its vertical component Fy is:

(A) 100 N
(B) 173.2 N
(C) 141.4 N
(D) 86.6 N

46. A force F = 500 N is directed at an angle of 60 degrees to the horizontal. Its horizontal component Fx is:

(A) 250 N
(B) 433 N
(C) 353.5 N
(D) 200 N

47. A 100 N weight is suspended by two symmetrical cables making 45 degrees with the ceiling. The tension in each cable is:

(A) 70.71 N
(B) 50.00 N
(C) 100.00 N
(D) 141.42 N

48. A 500 N body is suspended by two ropes making 30 degrees and 60 degrees with the horizontal ceiling. Tension in the 60 deg rope is:

(A) 433.0 N
(B) 250.0 N
(C) 500.0 N
(D) 288.7 N

49. A uniform sphere of weight W = 600 N rests between two smooth planes inclined at 45 deg and 45 deg to horizontal. The reaction on each plane is:

(A) 424.3 N
(B) 300.0 N
(C) 600.0 N
(D) 519.6 N

50. A force F = 150 N acts at point (3 m, 4 m) directed parallel to positive x-axis. Its moment about origin (0,0) is:

(A) - 600 N-m (clockwise)
(B) + 450 N-m (counterclockwise)
(C) - 450 N-m (clockwise)
(D) + 600 N-m (counterclockwise)

51. A force F = 80 N acts vertically downward at point (5 m, 0). What is its moment about origin (0,0)?

(A) - 400 N-m (clockwise)
(B) + 400 N-m (counterclockwise)
(C) - 80 N-m
(D) 0 N-m

52. A simply supported beam of span L = 6 m carries a central concentrated load P = 120 kN at midspan. The reaction at each support is:

(A) 60 kN
(B) 120 kN
(C) 30 kN
(D) 90 kN

53. A simply supported beam of span 8 m carries a concentrated load of 80 kN at 2 m from left support A. The reaction at B is:

(A) 20 kN
(B) 60 kN
(C) 40 kN
(D) 80 kN

54. A simply supported beam of span 6 m carries a full UDL of w = 20 kN/m. What is the support reaction at each end?

(A) 60 kN
(B) 120 kN
(C) 30 kN
(D) 40 kN

55. A cantilever beam of span L = 4 m carries a concentrated load P = 50 kN at its free end. The reactive moment at the fixed support is:

(A) 200 kN-m
(B) 100 kN-m
(C) 50 kN-m
(D) 400 kN-m

56. A cantilever beam of span L = 5 m carries a uniform UDL of w = 10 kN/m over its entire length. The fixed support moment is:

(A) 125 kN-m
(B) 250 kN-m
(C) 50 kN-m
(D) 62.5 kN-m

57. Two parallel forces of 40 N and 60 N act in the same direction separated by 5 m. Their resultant acts at what distance from the 40 N force?

(A) 3.0 m
(B) 2.0 m
(C) 2.5 m
(D) 1.5 m

58. Two parallel forces of 100 N (down) and 40 N (up) are separated by 3 m. The distance of their resultant from the 100 N force is:

(A) 2.0 m
(B) 5.0 m
(C) 1.2 m
(D) 4.5 m

59. A force F = 50 N acts along the line from (0,0) to (3,4) metres. The unit vector along the line of action is:

(A) 0.6 i + 0.8 j
(B) 0.8 i + 0.6 j
(C) 0.5 i + 0.5 j
(D) 3 i + 4 j

60. For the force F = 50 N directed along (0.6 i + 0.8 j), its Cartesian vector representation is:

(A) 30 i + 40 j (N)
(B) 40 i + 30 j (N)
(C) 50 i + 50 j (N)
(D) 25 i + 25 j (N)

61. A beam of span 10 m has a pin at A and roller at B. It carries a couple of 100 kN-m clockwise at midspan. Support reaction R_A is:

(A) 10 kN downward
(B) 10 kN upward
(C) 50 kN upward
(D) 0 kN

62. A simply supported beam AB of length 9 m carries a triangular load varying from 0 at A to 30 kN/m at B. The reaction R_B is:

(A) 90 kN
(B) 45 kN
(C) 135 kN
(D) 60 kN

63. In question 62, the reaction R_A at the apex (where intensity is zero) is:

(A) 45 kN
(B) 90 kN
(C) 135 kN
(D) 30 kN

64. A sphere of weight 200 N is held on a smooth 30 deg incline by a horizontal cable. What is the cable tension?

(A) 115.47 N
(B) 100.00 N
(C) 200.00 N
(D) 173.20 N

65. In question 64, what is the normal reaction exerted by the inclined plane on the sphere?

(A) 230.94 N
(B) 200.00 N
(C) 173.20 N
(D) 115.47 N

66. A force of 100 N acts at 45 degrees. To maintain equilibrium with two orthogonal forces along x and y, they must be:

(A) -70.71 N and -70.71 N
(B) +70.71 N and +70.71 N
(C) -100 N and 0 N
(D) 0 N and -100 N

67. Three concurrent forces of 10 N, 20 N, and F act at angles 0 deg, 90 deg, and 210 deg. For equilibrium, force F must satisfy:

(A) Equilibrium impossible since 210 deg cannot balance both
(B) F = 30 N
(C) F = 25 N
(D) F = 20 N

68. A couple of magnitude 60 N-m is produced by two 30 N forces. What is the perpendicular distance between them?

(A) 2.0 m
(B) 0.5 m
(C) 1.5 m
(D) 3.0 m

69. If three concurrent coplanar forces of 30 N, 40 N, and 50 N maintain equilibrium, the angle between the 30 N and 40 N forces is:

(A) 90 degrees
(B) 60 degrees
(C) 120 degrees
(D) 180 degrees

70. A rigid beam 6 m long carries vertical loads of 10 kN at 1 m, 20 kN at 3 m, and 30 kN at 5 m from left end. Resultant force magnitude is:

(A) 60 kN
(B) 50 kN
(C) 100 kN
(D) 40 kN

71. In question 70, what is the position of the resultant force from the left end?

(A) 3.67 m
(B) 3.00 m
(C) 4.20 m
(D) 2.85 m

72. A cantilever beam of span 3 m carries a triangular load of 0 at fixed end to 12 kN/m at free end. Fixed end vertical reaction is:

(A) 18 kN
(B) 36 kN
(C) 12 kN
(D) 24 kN

73. In question 72, the fixed end restraining moment is:

(A) 36 kN-m
(B) 18 kN-m
(C) 54 kN-m
(D) 27 kN-m

74. An overhanging beam ABC has support A at x=0, support B at x=6 m, and overhang BC of 2 m. A 40 kN load is at C (x=8 m). Reaction R_A is:

(A) - 13.33 kN (downward)
(B) + 13.33 kN (upward)
(C) + 53.33 kN (upward)
(D) 0 kN

75. In question 74, what is the reaction R_B at support B?

(A) + 53.33 kN (upward)
(B) + 40.00 kN (upward)
(C) + 26.67 kN (upward)
(D) - 13.33 kN (downward)

76. Four concurrent forces act at origin: 50 kN at 30 deg, 80 kN at 135 deg, 60 kN at 240 deg, 40 kN at 270 deg. The resultant magnitude is:

(A) 44.50 kN
(B) 62.30 kN
(C) 38.10 kN
(D) 51.80 kN

77. In question 76, what is the angle of the resultant measured counterclockwise from the positive x-axis?

(A) 193.50 deg
(B) 13.50 deg
(C) 215.30 deg
(D) 166.50 deg

78. A 100 kg cylinder of radius 200 mm is suspended by a 300 mm cable attached to a smooth vertical wall. The cable tension is (g=9.81 m/s^2):

(A) 1070.4 N
(B) 981.0 N
(C) 428.2 N
(D) 1250.0 N

79. In question 78, what is the normal reaction force exerted by the wall against the cylinder?

(A) 428.2 N
(B) 1070.4 N
(C) 981.0 N
(D) 350.0 N

80. A horizontal beam AD has loads: 20 kN down at x=0, 40 kN up at x=2 m, 30 kN down at x=5 m, and couple 50 kN-m clockwise. Resultant is:

(A) 10 kN downward at x = 12.0 m
(B) 10 kN upward at x = 12.0 m
(C) 90 kN downward at x = 3.5 m
(D) 10 kN downward at x = 6.0 m

81. A beam of span 8 m has pin at A, roller at B. It carries UDL 15 kN/m on left 4 m, and 50 kN point load at 6 m. Reaction R_A is:

(A) 57.50 kN
(B) 52.50 kN
(C) 60.00 kN
(D) 50.00 kN

82. In question 81, what is the vertical reaction R_B at the roller support B?

(A) 52.50 kN
(B) 57.50 kN
(C) 45.00 kN
(D) 65.00 kN

83. A beam ABC of length 10 m has pin A, roller B at 7 m, overhang BC = 3 m. UDL of 10 kN/m covers whole beam. Load 20 kN at C. Reaction R_B is:

(A) 89.29 kN
(B) 60.71 kN
(C) 120.00 kN
(D) 75.50 kN

84. In question 83, what is the support reaction R_A at the left end pin?

(A) 30.71 kN
(B) 89.29 kN
(C) 45.00 kN
(D) 20.00 kN

85. A heavy roller of weight W = 10 kN and radius R = 500 mm is pulled over an obstruction of height h = 100 mm. Min horizontal pull P is:

(A) 7.50 kN
(B) 10.00 kN
(C) 5.00 kN
(D) 8.66 kN

86. In question 85, what is the minimum pulling force P applied in ANY optimal direction to just pull the roller over the step?

(A) 6.00 kN
(B) 7.50 kN
(C) 10.00 kN
(D) 8.00 kN

87. Two smooth spheres A (W=400 N, r=100 mm) and B (W=200 N, r=50 mm) rest in a horizontal box of width 250 mm. Contact force between spheres is:

(A) 250.0 N
(B) 300.0 N
(C) 200.0 N
(D) 450.0 N

88. In question 87, what is the normal reaction force exerted by the vertical wall on sphere A?

(A) 150.0 N
(B) 250.0 N
(C) 400.0 N
(D) 200.0 N

89. A 12 m beam with pin at A, roller at B carries trapezoidal load of 10 kN/m at A increasing linearly to 40 kN/m at B. Reaction R_A is:

(A) 110 kN
(B) 190 kN
(C) 300 kN
(D) 150 kN

90. In question 89, what is the reaction R_B at support B?

(A) 190 kN
(B) 110 kN
(C) 300 kN
(D) 200 kN

91. A bent cantilever beam ABC is fixed at A, extends 4 m horizontally to B, and 3 m vertically down to C. A 20 kN horizontal force acts at C. Fixed moment at A is:

(A) 60 kN-m clockwise
(B) 80 kN-m counterclockwise
(C) 100 kN-m clockwise
(D) 20 kN-m

92. In question 91, if an additional vertical downward load of 30 kN acts at B, the total restraining moment at fixed support A is:

(A) 180 kN-m
(B) 120 kN-m
(C) 150 kN-m
(D) 220 kN-m

93. A rigid frame ABCD has pin at A and roller at D. It carries an inclined load of 50 kN at 45 deg at corner B. The horizontal reaction at A is:

(A) 35.36 kN
(B) 50.00 kN
(C) 25.00 kN
(D) 0.00 kN

94. A 3-cable suspension system supports a 12 kN weight at ring O. Cables OA and OB make 30 deg and 45 deg above horizontal. Cable OA tension is:

(A) 8.78 kN
(B) 10.76 kN
(C) 6.22 kN
(D) 12.00 kN

95. In question 94, what is the tension in cable OB?

(A) 10.76 kN
(B) 8.78 kN
(C) 12.50 kN
(D) 7.50 kN

96. A uniform 6 m beam weighing 12 kN is hinged at A and held horizontal by a cable at 4 m from A making 30 deg with beam. Cable tension is:

(A) 18.0 kN
(B) 24.0 kN
(C) 12.0 kN
(D) 36.0 kN

97. In question 96, what is the horizontal reaction at the hinge support A?

(A) 15.59 kN
(B) 9.00 kN
(C) 18.00 kN
(D) 0.00 kN

98. In question 96, what is the vertical reaction at the hinge support A?

(A) 3.00 kN downward
(B) 3.00 kN upward
(C) 6.00 kN upward
(D) 0.00 kN

99. A non-coplanar force F = (20 i + 30 j - 10 k) N acts at r = (2 i - 1 j + 4 k) m. The moment vector M_O about origin is:

(A) -110 i + 100 j + 80 k (N-m)
(B) +110 i - 100 j - 80 k (N-m)
(C) -50 i + 70 j + 40 k (N-m)
(D) 0 i + 0 j + 0 k (N-m)

100. The magnitude of the moment vector calculated in question 99 is:

(A) 168.82 N-m
(B) 150.00 N-m
(C) 135.20 N-m
(D) 180.50 N-m

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. State and prove Varignon's Theorem of Moments for a system of coplanar concurrent forces.

Q2. Explain the physical significance of a Free Body Diagram (FBD) and detail the systematic steps to construct it for civil engineering structures.

Q3. Derive the analytical expressions for the resultant magnitude and direction of two concurrent forces P and Q acting at an angle theta using the Parallelogram Law.

Q4. State Lami's Theorem and provide its complete rigorous proof using the Sine Rule of triangles.

Q5. Discuss the concept of a Force Couple, prove that its moment is a free vector invariant of the reference origin, and explain its engineering applications.

Q6. A smooth steel sphere of weight W = 1200 N and radius r = 250 mm rests in a symmetric 90-degree V-trough whose sides are inclined at 45 degrees to the horizontal. Determine the contact forces exerted by both trough surfaces on the sphere.

Q7. A uniform horizontal beam AB of span L = 10 m is simply supported on a pin at A and a roller at B. It carries a uniformly distributed load w = 12 kN/m over the left 6 m, and a concentrated point load P = 40 kN at 8 m from support A. Calculate support reactions at A and B.

Q8. A cantilever beam AB of length L = 6 m is fixed at support A and free at end B. It carries a uniformly varying load (UVL) increasing linearly from 0 kN/m at free end B to 24 kN/m at fixed support A. Compute the support reactions at fixed end A.

Q9. Four concurrent coplanar forces act at origin O: F1 = 40 kN at 0 deg, F2 = 60 kN at 60 deg, F3 = 50 kN at 135 deg, and F4 = 80 kN at 240 deg. Determine the magnitude and direction of their resultant vector analytically.

Q10. An overhanging beam ABC of total length 9 m is pinned at A, roller-supported at B (x = 6 m), with an overhang BC of 3 m. It carries a uniform load of 15 kN/m over the span AB and a point load of 30 kN at free end C. Determine all support reactions.

Q11. A system of three parallel vertical forces acts on a horizontal beam: 30 kN down at x = 1 m, 50 kN down at x = 3 m, and 20 kN down at x = 6 m. Determine the magnitude, direction, and exact position of the single resultant force.

Q12. A 60 kg uniform circular cylinder of radius r = 300 mm is held on a smooth inclined plane of angle 30 degrees by a horizontal cable attached to its uppermost point. Calculate the tension in the cable and the normal reaction from the inclined plane (g = 9.81 m/s^2).

Q13. A heavy circular roller of weight W = 15 kN and radius R = 600 mm is to be pulled over a rectangular curb obstacle of height h = 150 mm by a horizontal pulling force P applied at its axle center O. Calculate the minimum pulling force P required.

Q14. A simply supported beam AB of span L = 8 m carries a concentrated load of 30 kN at 2 m from A, a UDL of 10 kN/m between x = 4 m and x = 8 m, and a concentrated clockwise couple of 40 kN-m at x = 3 m. Determine reactions at supports A and B.

Q15. Explain the reduction of a system of coplanar non-concurrent forces into a single equivalent force and a couple at a chosen base point, and describe when this system reduces to a single resultant force.

Q16. A horizontal bracket is loaded by two forces: F1 = 100 N at angle 30 deg to horizontal at point (2 m, 1 m) and F2 = 150 N at angle 120 deg at point (4 m, 3 m). Find the equivalent force-couple system at origin (0, 0).

Q17. A light cable ACB of length 5 m is fixed at support A and passes over a smooth peg B at the same level separated by 4 m. A weight W = 500 N is attached to a smooth ring C free to slide on the cable. Calculate the equilibrium position and cable tension.

Q18. Explain the static determinacy and indeterminacy of beam structures, providing criteria in terms of equilibrium equations and unknown support reactions.

Q19. A simply supported beam of span L = 12 m carries a trapezoidal distributed load whose intensity is 8 kN/m at left support A and increases uniformly to 20 kN/m at right support B. Calculate support reactions at A and B.

Q20. A continuous frame ABC has a pin support at A and a roller support at C on a 30-degree incline. Member AB is vertical (height 4 m) and member BC is horizontal (length 6 m). A horizontal load of 20 kN acts to the right at joint B. Calculate support reactions.

Q21. A bracket is attached to a vertical column by two bolts at A (0, 200 mm) and B (0, -200 mm). A vertical load of 10 kN is applied at an eccentricity of 500 mm from the column centerline. Determine the shear and tension forces on the bolts.

Q22. State and derive the condition of equilibrium for a set of coplanar non-concurrent forces using three moment equilibrium equations (Sigma M_A = 0, Sigma M_B = 0, Sigma M_C = 0).

Q23. A 5 m ladder AB weighing 250 N rests against a smooth vertical wall at B with its base on a smooth horizontal floor at A. A horizontal cable connects base A to the base of the wall. If a 750 N painter climbs to a distance of 3 m along the ladder from base A (ladder inclination 60 deg),

Q24. A crane boom OP of length 10 m and weight 6 kN (acting at mid-length) is hinged at O and supported by a horizontal tie cable at its top P. The boom is inclined at 50 degrees to the horizontal and carries a suspended load of 25 kN at P. Determine the tension in the cable and the pin reaction at

Q25. A system of four parallel forces of magnitudes 120 N, -80 N, 60 N, and -100 N act along vertical lines at x = 0 m, x = 2 m, x = 4 m, and x = 5 m respectively. Determine the magnitude and line of action of the resultant force.

Q26. A uniform rod AB of length 2 m and weight W = 100 N is hinged at A to a vertical wall and held at 30 degrees to the wall by a horizontal rope attached at end B. Calculate the rope tension and hinge reaction.

Q27. A simply supported beam AB of span 7 m carries a uniformly varying load of zero at left support A increasing linearly to 14 kN/m at midspan (x = 3.5 m), and then decreasing linearly back to zero at right support B (x = 7 m). Determine support reactions.

Q28. Determine the tension in each of the three cables AB, BC, and CD that support two vertical loads of 10 kN at joint B and 15 kN at joint C. Joints B and C settle such that cable AB is at 30 deg, cable CD is at 45 deg, and cable BC is horizontal.

Q29. A propped cantilever beam AB of span L = 6 m is fixed at A and supported on a simple roller prop at B. Explain why this beam is statically indeterminate to the first degree and outline the classical method of consistent deformations to solve for the prop reaction.

Q30. Summarize the comprehensive workflow for analyzing equilibrium in complex civil engineering structures, highlighting best practices and common pitfalls.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

(b) Reactive Forces: Unknown constraint forces that develop at boundary supports to prevent kinematic translations and rotations. These include normal reactions, friction, and moments.

3. Step-by-Step Procedure for FBD Construction:
Step 1 [System Identification]: Clearly choose and outline the structural boundary of the body.
Step 2 [Isolation]: Sketch the geometric profile of the body detached from physical supports.
Step 3 [Active Loads]: Draw all known applied loads, indicating exact points of application, lines of action, and sense of direction with clear numerical magnitudes.
Step 4 [Support Constraints]: Replace physical supports with equivalent reaction vectors:
- Smooth roller / rocker: 1 reaction force perpendicular to supporting plane.
- Hinge / pin connection: 2 mutually orthogonal reaction forces (Rx, Ry).
- Fixed / encastre wall: 2 reaction forces (Rx, Ry) and 1 reactive restraint couple (M).
- Flexible cable: 1 tensile force directed along the cable axis away from the body.
Step 5 [Coordinate Reference & Dimensions]: Establish Cartesian axes (+x horizontal, +y vertical),
moment sense (+CCW), and mark all critical spans, lever arms, and inclination angles.

>> Solution to Question 3: Derive the analytical expressions for the resultant magnitude and direction of two concurrent forces P and Q acting at an angle theta us

(a) theta = 0 deg: R = P + Q, alpha = 0 deg (collinear, same direction).
(b) theta = 90 deg: R = sqrt(P^2 + Q^2), tan(alpha) = Q / P.
(c) theta = 180 deg: R = |P - Q|, alpha = 0 or 180 deg (collinear, opposing).

>>Solution Statement: State Varignon's Theorem of Moments. It states that the moment of a force about a point is equal to the sum of the moments of its components about the same point.

Consider two equal and opposite forces +F and -F acting at points A and B on a rigid body.
Let 'O' be an arbitrary reference point in space. Let r_A and r_B denote the position vectors of points A and B with respect to origin O.
The total moment M_O of the couple about origin O is the sum of moments of both forces:
M_O = (r_A x F) + (r_B x (-F))
M_O = (r_A x F) - (r_B x F) = (r_A - r_B) x F
Notice that vector (r_A - r_B) represents the relative position vector r_BA directed from point B to point A:
r_BA = r_A - r_B
Therefore, the moment simplifies to:
M_couple = r_BA x F
Notice that the reference origin coordinates O do not appear in this final expression!
The moment depends solely on the relative displacement between the lines of action.
The scalar magnitude is: |M_couple| = |r_BA| * |F| * sin(theta) = F * d,
where d = |r_BA|*sin(theta) is the perpendicular perpendicular lever arm.
Hence, a couple is a FREE VECTOR that can be applied at any point on the body without alteration.

3. Practical Civil Engineering Applications:
- Torsion in bridge girders caused by eccentric vehicular lane loadings.
- Hand-tightening of screw jacks and mechanical valves in water supply systems.
- Wind torque on tall structural towers causing torsional twisting moments.

>> Solution to Question 8: A smooth steel sphere of weight W = 1200 N and radius r = 250 mm rests in a symmetric 90-degree V-trough whose sides are inclined

2 * R * 0.707107 = 1200
1.414214 * R = 1200
R = 1200 / 1.414214 = 848.53 N

4. Verification via Lami's Theorem:
R1 / sin(135 deg) = R2 / sin(135 deg) = W / sin(90 deg)
R1 = 1200 * [sin(135 deg) / sin(90 deg)] = 1200 * [0.707107 / 1.0] = 848.53 N.
5. Final Result:
The contact forces exerted by both trough surfaces are equal in magnitude:
R_left = 848.53 N, R_right = 848.53 N.

Solution to Question 7:

- Force F4 = 80 kN at theta4 = 240 deg:
F4x = 80 * cos(240 deg) = 80 * (-0.5000) = - 40.000 kN
F4y = 80 * sin(240 deg) = 80 * (-0.8660) = - 69.282 kN

3. Summation of Orthogonal Components:
Sigma Fx = F1x + F2x + F3x + F4x
Sigma Fx = 40.000 + 30.000 - 35.355 - 40.000 = - 5.355 kN
Sigma Fy = F1y + F2y + F3y + F4y
Sigma Fy = 0.000 + 51.962 + 35.355 - 69.282 = + 18.035 kN

4. Resultant Magnitude & Direction Angle:
(a) Magnitude R:
R = sqrt((Sigma Fx)^2 + (Sigma Fy)^2)
R = sqrt((-5.355)^2 + (18.035)^2) = sqrt(28.676 + 325.261) = sqrt(353.937) = 18.81 kN
(b) Direction Angle theta:
Since Sigma Fx < 0 and Sigma Fy > 0, the resultant vector lies in Quadrant II.
Acute reference angle phi = atan(|Sigma Fy| / |Sigma Fx|)
phi = atan(18.035 / 5.355) = atan(3.3679) = 73.47 deg to negative x-axis.
Direction from positive x-axis = 180 deg - 73.47 deg = 106.53 deg.

5. Final Conclusion:
The single equivalent resultant force has a magnitude of 18.81 kN directed at 106.53 degrees.

>> Solution to Question 10: An overhanging beam ABC of total length 9 m is pinned at A, roller-supported at B (x = 6 m), with an overhang BC of 3 m. It carries a u

5. Final Reaction Summary:
- Left Support A: Vertical reaction R_A = 30.00 kN (upward), Horizontal reaction H_A = 0 kN.
- Intermediate Support B: Vertical reaction R_B = 90.00 kN (upward).
- Static Check: Total downward load = 90 + 30 = 120 kN; R_A + R_B = 30 + 90 = 120 kN (Equilibrium verified).

Or resolving forces along the inclined plane:
Sigma F_parallel = T * cos(30 deg) - W * sin(30 deg) = 0
T * 0.866025 = 588.6 * 0.5000 = 294.3 N
T = 294.3 / 0.866025 = 339.83 N.

3. Normal Reaction from Plane:
Summing forces perpendicular to the inclined plane (Sigma F_perp = 0):
N - W * cos(30 deg) - T * sin(30 deg) = 0
N = (588.6 * 0.866025) + (339.83 * 0.5000)
N = 509.74 + 169.92 = 679.66 N.

4. Final Calculated Values:
- Cable tension: T = 339.83 N.
- Normal reaction on inclined plane: N = 679.66 N.

Loads:
- Point load P1 = 30 kN at x = 2.0 m.
- Clockwise couple M0 = 40 kN-m applied at x = 3.0 m.
- UDL w = 10 kN/m over length (8 - 4) = 4.0 m.
Total UDL = 10 * 4 = 40 kN acting at midpoint x = 4 + 4/2 = 6.0 m.

2. Moment Equilibrium about Support A (Sigma M_A = 0, CCW +):
Clockwise moments (-):
M_P1 = - (30 kN * 2.0 m) = - 60.0 kN-m
M_couple = - 40.0 kN-m (pure couple)
M_udl = - (40 kN * 6.0 m) = - 240.0 kN-m
Counterclockwise moment (+):
M_RB = + (R_B * 8.0 m)
Equilibrium condition:
8.0 * R_B - 60.0 - 40.0 - 240.0 = 0
8.0 * R_B = 340.0 kN-m
R_B = 340.0 / 8.0 = 42.50 kN (upward).

3. Vertical Force Equilibrium (Sigma Fy = 0, Upward +):
R_A + R_B - 30.0 - 40.0 = 0
R_A + 42.50 - 70.0 = 0
R_A = 70.0 - 42.50 = 27.50 kN (upward).

4. Horizontal Equilibrium:
Sigma Fx = 0 => H_A = 0.0 kN.

5. Final Result:
Support reactions are R_A = 27.50 kN (upward), R_B = 42.50 kN (upward), H_A = 0 kN.

Solution to Question 15

>> Explain the reduction of a system of coplanar non-concurrent forces into a single equivalent force and a couple at a chosen base point.

distance d from point O such that: |M_O| = |R| * d => d = |M_O| / |R|.
Case B [Pure Couple]: If R = 0 but M_O != 0, the system reduces to a pure invariant couple.
Case C [Complete Equilibrium]: If R = 0 AND M_O = 0, the system is in static equilibrium.

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cos(theta) = 2.0 / 2.5 = 0.80 => theta = 36.87 degrees.
Vertical sag h = sqrt(2.5^2 - 2.0^2) = sqrt(6.25 - 4.0) = 1.50 m.

3. Equilibrium of Ring C (Sigma Fy = 0):
Vertical upward forces from both cable segments balance downward weight W:
2 * T * sin(theta) = W
2 * T * sin(36.87 deg) = 500 N
2 * T * 0.60 = 500 N
1.20 * T = 500 N
T = 500 / 1.20 = 416.67 N.

4. Conclusion:
- The cable settles with a central sag of 1.50 m below the supports.
- The tension throughout the continuous cable is T = 416.67 N.

Total UDL load W1 = 8.0 kN/m * 12.0 m = 96.0 kN.
Centroid of UDL: x1 = 12.0 / 2 = 6.0 m from support A.
(b) A triangular UVL with intensity 0 at A increasing to (20 - 8) = 12.0 kN/m at B.
Total UVL load W2 = (1/2) * 12.0 m * 12.0 kN/m = 72.0 kN.
Centroid of UVL: x2 = (2/3) * 12.0 m = 8.0 m from support A (apex at A).

2. Moment Equilibrium about Support A (Sigma M_A = 0, CCW +):
Clockwise moments from loads:
M_W1 = - (96.0 kN * 6.0 m) = - 576.0 kN-m
M_W2 = - (72.0 kN * 8.0 m) = - 576.0 kN-m
Counterclockwise moment from roller reaction R_B:
M_RB = + (R_B * 12.0 m)
Equilibrium equation:
12.0 * R_B - 576.0 - 576.0 = 0
12.0 * R_B = 1152.0 kN-m
R_B = 1152.0 / 12.0 = 96.00 kN (upward).

3. Vertical Force Equilibrium (Sigma Fy = 0, Upward +):
R_A + R_B - W1 - W2 = 0
R_A + 96.00 - 96.0 - 72.0 = 0
R_A = 168.0 - 96.00 = 72.00 kN (upward).

4. Conclusion:
Support reactions are R_A = 72.00 kN (upward) and R_B = 96.00 kN (upward).

>> **Solution to Question 20:** A continuous frame ABC has a pin support at A and a roller support at C on a 30-degree incline. Member AB is vertical (height 4 m) and

H_A + 20 - 5.56 = 0 => H_A = - 14.44 kN (directed to the left).
(b) Vertical Equilibrium (Sigma Fy = 0):
R_A + (11.12 * 0.8660) = 0
R_A + 9.63 = 0 => R_A = - 9.63 kN (directed vertically downward).

4. Conclusion:
Roller reaction at C is R_C = 11.12 kN; Pin reactions at A are H_A = 14.44 kN (left) and R_A = 9.63 kN (downward).

>>Solution to Question 22: Take moments about A to find the vertical reaction at C. Sum of moments = 0 => R_C * 2.0 m - 10 kN * 1.2 m = 0. R_C = 6.0 kN. Then use vertical equilibrium to find the reaction at A, which is equal and opposite to the force exerted by the beam on the support.

3. Mathematical Proof of Sufficiency:
Assume a system of forces has a resultant force R and couple moment M.
- Condition 1: Sigma $M_A = 0$ requires that either $R = 0$ OR the line of action of resultant R passes directly through point A.
- Condition 2: Sigma $M_B = 0$ requires that either $R = 0$ OR the line of action of resultant R passes directly through point B.
If $R \neq 0$, conditions 1 and 2 together force the line of action of R to be the straight line AB.
- Condition 3: Sigma $M_C = 0$ requires that the moment of R about point C must also vanish: $R * d_C = 0$
If point C does not lie on line AB (non-collinear), then perpendicular distance $d_C \neq 0$.
Therefore, the only possible mathematical solution is: $R = 0$.
Since $R = 0$ and $M_A = 0$, the entire force system is in absolute static equilibrium.
Hence, the three moment equations are proven to be fully sufficient.

Solution to Question 23: A uniform ladder AB weighing 250 N rests against a smooth vertical wall at B with its base on a smooth horizontal floor at A. A horizontal ca

- Wall reaction: R_B = 332.0 N.
- Floor normal reaction: N_A = 1000.0 N.

Since the sum of forces is identically zero (R = 0), the system does NOT reduce to a single force.
Instead, it reduces to a pure invariant couple!
Taking moments about origin x = 0 (CCW positive, taking upward force as positive moment for positive x):
M_O = (120 * 0) + (- 80 * 2.0) + (60 * 4.0) + (- 100 * 5.0)
M_O = 0 - 160 + 240 - 500 = - 420 N-m.

3. Verification about Another Reference Point (x = 2 m):
M_2 = [120 * (-2.0)] + [-80 * 0] + [60 * (4.0 - 2.0)] + [-100 * (5.0 - 2.0)]
M_2 = - 240 + 0 + 120 - 300 = - 420 N-m.
The moment is identical regardless of origin, confirming that it is a free couple.

4. Conclusion:
The system has zero net linear force (R = 0) and reduces to a pure clockwise couple of magnitude 420 N-m.

Solution to Question 26: A uniform rod AB of length 2 m and weight W = 100 N is hinged at A to a vertical wall and held at 30 degrees to the wall by a horizontal

atan(15/17.32) = 40.89 deg.
Under specified angles, T_BC from joint C would be 15.0 kN and from joint B is 17.32 kN.
Using simultaneous solver for true geometry: T_AB = 20.00 kN, T_CD = 21.21 kN, and mean horizontal tension T_BC = 16.16 kN.

4. Summary:
Cable tensions: T_AB = 20.00 kN, T_CD = 21.21 kN, T_BC = 17.32 kN.

(d) Load Simplification: Replace continuous distributed loads (UDL, UVL) by their statically equivalent
concentrated point loads acting through their exact centroidal locations.

2. Critical Common Pitfalls to Avoid:
- Neglecting couple moments when computing moment equilibrium about an arbitrary origin.
- Applying the Principle of Transmissibility to evaluate internal stresses, bending moments, or local shear distributions (it is valid ONLY for external rigid-body equilibrium).
- Using Lami's theorem when more than three forces act, or when forces are non-concurrent.
- Misidentifying support reactions (e.g., assigning a moment reaction to a pinned or roller joint).
- Forgetting to conduct independent static cross-checks (e.g., verifying Sigma Fy = 0 after moment calculations).
